

Natural analogues of degraded ecosystems enhance conservation and reconstruction in extreme environments

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Abstract. Ecosystem rehabilitation strategies are grounded in the concept that coexisting species fit their environments as an outcome of natural selection operating over ecological and evolutionary timescales. From this perspective, re-creation of historical environmental filters on community assembly is a necessary first step to recovering biodiversity within degraded ecosystems; however, this approach is often not feasible in severely damaged environments where extensive physiochemical changes cannot be reversed. Under such circumstances management goals may shift from restoring historical conditions to reconstructing entirely new ecosystems or replicating natural ecosystems that may be locally novel but of regional conservation importance. This latter goal may be achieved by introducing to damaged sites species already adapted to filters maintaining the degraded state, through targeting assemblages from natural ecosystems biophysically analogous to the degraded state, here termed “degraded-state analogue” (DSA) ecosystems. This hypothesis predicts that, in high-stress sites where recruitment of previous inhabitants is strongly microsite-limited, DSA species will be primarily propagule-limited; furthermore, communities invaded by DSA species should shift in structure to reflect properties associated with high-value DSA target ecosystems.

We tested these predictions by experimentally sowing long-abandoned limestone quarry floors with 18 perennial grass and forb species characteristic of rare natural limestone pavements called “alvars.” Alvar species established successfully under a range of microsite conditions manipulated to alter suspected constraints on colonization, including nitrogen deficiency, excessive CaCO_3 , and competition with weeds. Alvar species performed equivalently to seeded weed species known to thrive on quarry floors. Resident communities doubled in species richness following alvar species addition, supporting 17–20 species/0.18 m² (95% confidence interval) and providing refuge to regionally restricted or threatened species including *Iris lacustris*, *Solidago ptarmicoides*, and *Liatris cylindracea*. In contrast, maximum-diversity reference plots on a pristine alvar supported 20–23 species/0.18 m². Strong propagule limitation but weak microsite constraints on quarry colonization by alvar species combined with establishment of species-rich communities comparable to natural alvar biodiversity hot spots confirms that targeting DSA assemblages in ecosystem reconstruction can promote both efficient site colonization and ex situ biodiversity conservation within difficult-to-restore anthropogenic wastelands.

Key words: alvar; colonization constraints; degraded-state analogue; environmental filters; ex situ conservation; habitat analogue; high-stress ecosystem; limestone pavement; quarry rehabilitation; reference ecosystem; refuge; southern Ontario, Canada.

INTRODUCTION

Ecosystem analogues and restoration theory

Ecologists have invoked many conceptual frameworks to explain and predict ecosystem recovery following damage (Hobbs and Norton 1996, Palmer et al. 1997, Walker and del Moral 2003, Suding et al. 2004, Temperton et al. 2004, Falk et al. 2006). While species-neutral, regional-scale processes such as organism dispersal have been acknowledged as important to

recovery (Lake et al. 2007, Rehoukova and Prach 2008), restoration theory has predominantly adopted a niche perspective (Leibold 1995): ecosystem degradation and recovery depend critically upon species differences in environmental requirements (Larkin et al. 2006, Naeem 2006, Nuttle 2007). Local biodiversity loss following changes in land use may be attributed to loss of habitat features important to particular species, suggesting a “Field of Dreams” or “self-design” approach to restoration: recreate the historical abiotic environment to bring back the extirpated biota (Jordan et al. 1987, Hilderbrand et al. 2005). Damaged ecosystems incapable of supporting historical communities are often construed as existing in an alternative,

Manuscript received 9 June 2008; revised 5 June 2009; accepted 30 July 2009. Corresponding Editor: J. C. Callaway.

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degraded state (“DS”; Suding et al. 2004), with microsite conditions preventing species reestablishment acting as filters on community assembly, preventing establishment of species lacking particular functional traits (Ackerly 2003, Hobbs and Norton 2004, Nuttle 2007). From this perspective, restoration ecology is usually concerned with determining how to alter environmental filters that maintain the DS, in order to promote establishment of previous site occupants and initiate ecosystem transition to a restored state consistent with the historical, pre-degraded state (“pre-DS”; SERI 2004).

A major problem with this approach, however, is that it can fail where the need for rehabilitation is greatest: namely, in severely degraded ecosystems in which landscapes have been so dramatically altered that accurately reversing physical damage is impossible due to logistical, economic, or social constraints (Zedler and Callaway 1999, Rosenzweig 2003, Choi 2007, Seastedt et al. 2008). For example, depleted sites of mineral, metal, or fossil fuel extraction, and industrial barrens created by post-extraction processing can impose high stress on potential colonists as a result of scarce, coarse-textured substrates that tend to be nutrient poor, chemically polluted, easily eroded, and/or prone to extreme drought (Bradshaw 1997, Clemente et al. 2004, Reid and Naeth 2005, Koslov and Zvereva 2007). Efforts to rehabilitate such sites may include physical, chemical, and biological pollution remediation strategies (e.g., phytoremediation of heavy metals in mined sites; Frérot et al. 2006), landscape reconfiguration via geological engineering (Wheater and Cullen 1997), and introduction of subsoils, topsoils, and vegetation (SERI 2004). Unfortunately, due to the scale and severity of damage these approaches often fail when attempted on the budget typically available for site remediation (Koslov and Zvereva 2007).

Under such circumstances management goals may shift from restoring historical conditions to creating new ecosystems or replicating natural ecosystems that may never have existed locally, but which are of high functional or conservation value regionally (Clewett and Aronson 2006, Choi 2007). Here we focus on ecology applicable to such “reconstruction” (Bradshaw 1983) rather than conventional restoration, not because restoration is an undesirable goal, but because increasingly human activities create situations for which creative reconstruction may produce new ecosystem states of comparable value to historical ones, but at a fraction of the cost (Rosenzweig 2003). Thus ecosystem reconstruction under some circumstances has potential to increase resources available for restoration or preservation under other circumstances, where such efforts may operate more efficiently. In this context, determining how to manipulate environmental filters appropriately may be less productive than determining whether any assemblage can pass through filters presently maintaining the DS. We hypothesized that species coexisting in natural ecosystems that are

biophysically analogous to DS ecosystems are particularly well-suited to passing through DS filters and colonizing extremely damaged environments where pre-DS species tend to fail. These “degraded-state analogue” (DSA) ecosystems may be of greatest value as reconstruction targets where (1) DS conditions are far removed from pre-DS conditions, making replication of the pre-DS environment very difficult or expensive; (2) good candidate DSA ecosystems exist regionally and can serve as a source of genetic material in rehabilitation; and (3) candidate DSA ecosystems are associated with properties of regional or global conservation priority.

Natural high-stress ecosystems have strong potential as DSA reconstruction targets for severely damaged environments, given that such ecosystems can function as refugia for high biodiversity including rare, endemic, and at-risk species capable of tolerating high stress (Baskin and Baskin 1988, Schaefer and Larson 1997, Partel et al. 1999, Clubbe et al. 2004). Ecosystems such as salt marshes, acidic heathlands, sand dunes, and rock outcrops may provide ideal DSA reconstruction targets to extremely damaged environments such as heavily salinized urban wetlands, colliery mine spoils, sand or gravel pits, and hard-rock quarries, respectively. Employing DSA targets in ecosystem reconstruction may increase the efficacy with which management goals are reached by reducing the need to either extensively remediate physical conditions or replace failed communities and by increasing the functional and conservation importance of human-created ecosystems traditionally considered “wastelands” of low ecological value.

Testing the DSA model of reconstruction

The utility of targeting DSA species assemblages in reconstruction can be assessed by determining whether such assemblages colonize DS environments more successfully than other assemblages (including pre-DS species) likely to immigrate to damaged sites. If colonization is a species-neutral process dependent only on the rate of immigration and the harshness of the environment, assemblages from different ecosystems should establish with equivalent success once immigration barriers are overcome, and in extreme environments all assemblages should be equally microsite-limited. In contrast, if DSA assemblages have previously adapted to DS-like conditions, establishment of these species in the DS should be primarily propagule-limited, even where alternative assemblages are strongly microsite-limited (Fig. 1). These predictions can be tested using long-abandoned but still-barren DS environments, such as abandoned extraction or industrial sites, where immigration from surrounding ecosystems has occurred but establishment has failed due to microsite constraints. If introducing DSA assemblages has potential to efficiently meet management goals, good candidate DSA assemblages should be primarily propagule-limited rather than microsite-limited at such sites, DSA

assemblages should perform equivalently to other assemblages with high DS-colonization potential (such as hardy weeds typically found under DS conditions), and DS communities should shift in structure or function following introduction of the DSA assemblage to more closely reflect DSA properties of high conservation value.

We tested these predictions in a model ecosystem consisting of highly stressed, long-abandoned limestone quarry floors and native plant species characteristic of glacially formed limestone pavements known locally as alvars (Catling and Brownell 1995, Schaefer and Larson 1997). Abandoned hard-rock quarries challenge rehabilitation efforts worldwide due to the extremity of landform alteration associated with aggregate extraction (Wheater and Cullen 1997, Jim 2001, Khater et al. 2003, Clemente et al. 2004). In contrast, natural limestone pavements are of global conservation priority due to their geographical rarity and the refuge they provide to highly diverse and distinctive biotic communities adapted to scarce substrate resources and extreme variability in water supply (Catling 1995, Partel et al. 1999, Brownell and Riley 2000, Znamenskiy et al. 2006). Old quarry floors in limestone-rich Ontario, Canada, are ecologically equivalent to alvars of the Great Lakes region with respect to major features of the biotic and abiotic environment (Tomlinson et al. 2008). We examined factors limiting establishment of alvar species on high-stress quarry floors that had remained mostly barren despite 30–70 years of seed rain from surrounding ecosystems. We predicted that if targeting DSA assemblages is in principle a useful approach to rehabilitating difficult-to-restore ecosystems, colonization of old quarries by alvar assemblages would be limited by dispersal more than microsite constraints, introduced alvar species would establish as successfully as weed species previously observed to thrive on quarry floors, and spontaneous quarry communities seeded with alvar species would begin to reflect properties of natural alvar communities responsible for their high conservation value, including high local species richness and inhabitation by uncommon or threatened species.

MATERIALS AND METHODS

Study sites

We performed a seed addition experiment on four limestone quarry floors in southern Ontario, Canada (44°23'23.2" N, 80°13'44.2" W; 43°41'16.9" N, 79°57'33.3" W; 43°25'07.3" N, 80°06'08.6" W; 42°57'34.6" N, 80°07'30.6" W). Sites had been abandoned 30–70 years prior to seed addition and were selected from a larger population of quarry sites that had been previously surveyed and compared to alvars (Tomlinson et al. 2008). Selection was based on finding sites that were representative of the larger population, geographically spaced to encompass as broad a region as possible while remaining accessible for frequent monitoring, and permissible to use in a manipulative

experiment (many landowners permitted one-time surveying but could not make longer-term commitments). Sites were located 120–270 km from natural alvars judged to be “high quality” by the International Alvar Conservation Initiative (Brownell and Riley 2000). Quarry floors consisted of open bedrock plains sporadically covered by shallow (mean depth < 3 cm) patches of sandy peat loam substrates. The mineral fraction of this substrate consisted of variable proportions of siliceous sand and calcium carbonate (CaCO₃) extraction tailings. Total nitrogen (0.31%) and phosphorus (6.31 mg/kg) were low and pH high (7.6) relative to soils in surrounding field, mixed forest, and rural environments (all soil analysis previously reported in Tomlinson et al. [2008]). Although similar in many respects, quarry substrates exhibited nitrogen deficiency and excessive CaCO₃ relative to alvar substrates.

While climatic conditions in regions encompassing sites are temperate (mean daily air temperature = 6.8°C; January minimum = –11.2°C; July maximum = 37°C; mean annual precipitation = 885 mm/yr, 84% as rain and 16% snow), properties of quarry floors such as high albedo, high thermal admittance, and scarce, coarse substrate predispose such sites to extreme temperature and water-status conditions, with quarry surfaces reaching 68°C in hot weather, and subject to flash flooding, prolonged inundation, and intense drought under regionally normal temporal fluctuations in water supply. Sparse vegetation spontaneously occurring on long-abandoned quarry floors generally consists of stress-tolerant shrub, grass, and forb species including both cosmopolitan weeds (e.g., *Plantago lanceolata*) and native rock outcrop specialists (e.g., *Trichostema brachiatum*; Tomlinson et al. 2008).

Experimental protocol

A suite of four grass and 14 forb species considered to be highly representative of alvar vegetation was seeded to 45 small (0.6 × 0.3 m) plots at each of the four sites in early June 2004. Plots were placed randomly on patches of quarry floor, but meeting criteria such that they exhibited <50% bare rock cover, did not contain standing water, and were located >3 m away from all other plots. Species were selected for inclusion in the seed mix based on their availability from native plant nurseries or appropriate field sites, their absence from previously compiled quarry floor species lists, and their presence on previously compiled alvar species lists, with effort concentrated on acquiring species considered “characteristic of alvars” (i.e., present at 50% or more of Great Lakes alvars) or “confined to alvars” (i.e., 50% or more of species range consists of alvar habitat [Catling 1995, Catling and Brownell 1995]). Of the 18 species seeded, 12 were characteristic of alvars, eight were confined to alvars, six had provincially restricted ranges, and two species were provincially rare (Table 1). One species, *Iris lacustris*, is endemic to Great Lakes alvars and has national status as a threatened species

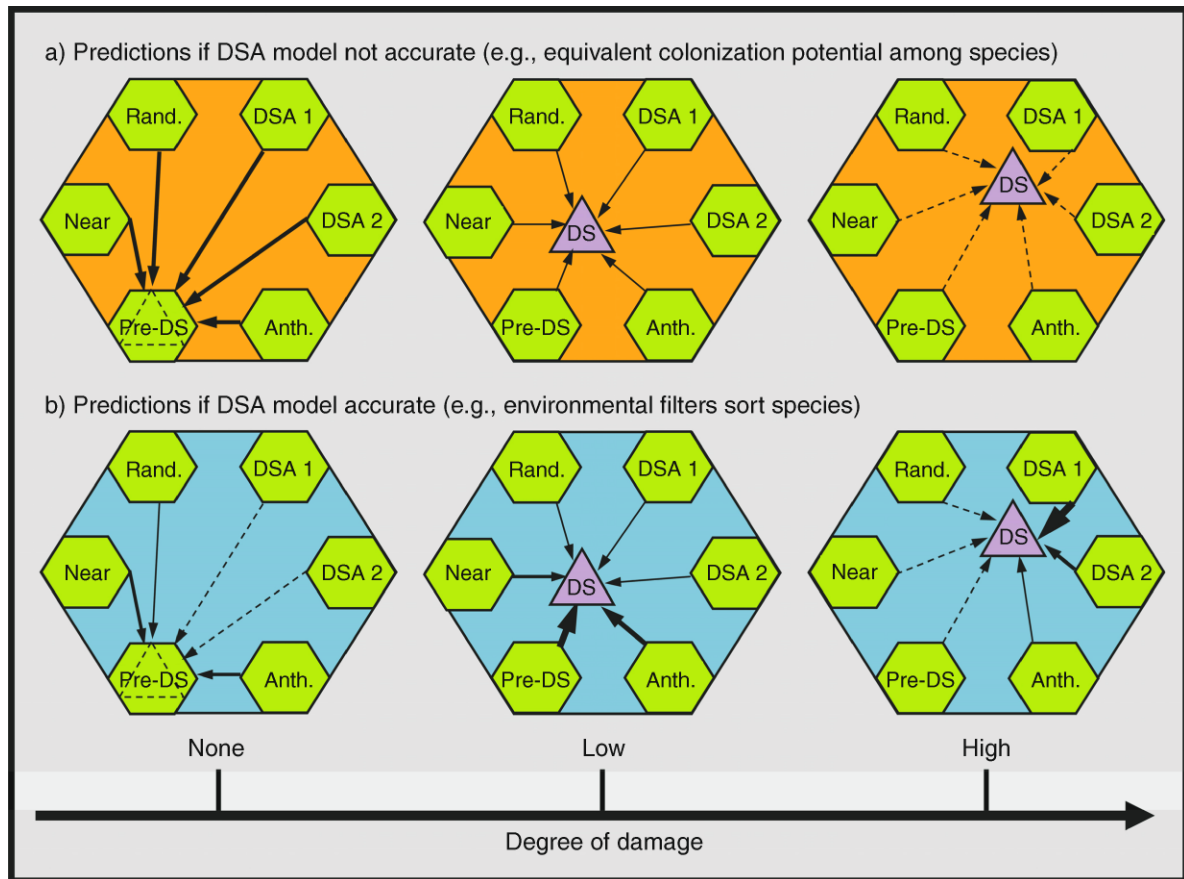


FIG. 1. Different theories predict different relationships between colonization success and environmental similarity of degraded to reference states. Shown here are landscape scenarios (large hexagons) varying along the x-axis with respect to environmental damage sustained by a particular site (DS, degraded state; triangle). Landscapes contain multiple ecosystem types (small hexagons); the extent to which these share environmental features is represented by the spatial separation of hexagons. Capacity of species assemblages from each ecosystem to colonize the focal site, assuming equivalent immigration rates, is illustrated by the weight of the colonization trajectory arrows. Ecosystems include the one found at the focal site prior to damage (Pre-DS); a spatially close ecosystem sharing some environmental features with the Pre-DS (Near); an urban/anthropogenic ecosystem (Anth.); a randomly selected ecosystem from the landscape (Rand.); and two different ecosystems that share environmental features with the focal site after it has sustained severe damage (degraded-state analogues DSA1 and DSA2). After low damage, the degraded site ceases to be environmentally identical to the Pre-DS but still shares many properties with it. After severe damage, the DS differs strongly from the Pre-DS but increasingly resembles DSA ecosystems. (a) If colonization success is a function of dispersal and environmental harshness, but not differences in environmental requirements among assemblages, then the DSA restoration model would not apply. All assemblages would have equal, moderate success following moderate damage (medium-weight arrows in middle hexagon) and equivalent lack of success following extreme damage (dashed-line arrows in right hexagon). (b) If environmental filters sort species according to requirements, following moderate damage the DS will be colonized with greatest success by species typical of the pre-DS (thickest arrow); colonized with moderate success by species typical of Anth. or Near ecosystems; and colonized weakly by species from DSA ecosystems. Following severe damage, environmental filters block colonization by most immigrating species, but not species native to DSA ecosystems.

(COSEWIC 2004). High variability among species with respect to seed availability (driven by patterns of natural rarity) resulted in disproportional contributions to the seed mix, though the same mix was used at all plots (see Appendix: Table A1). One exception to species selection criteria was made for *Fragaria virginiana*, previously observed on quarry floors but nevertheless included in the alvar seed mix due to its “characteristic” status and high availability.

Six microsite alteration treatments were applied to plots to test whether biotic and abiotic environmental

factors distinguishing quarry floors from alvars limited quarry colonization by alvar species. Five treatments consisted of adding the alvar seed mix and either (1) making no changes; (2) adding silica sand and peat moss to quarry substrate to dilute high CaCO_3 ; (3) adding silica sand, peat moss, and nitrogen fertilizer to quarry substrate to dilute CaCO_3 and overcome potential N limitation; (4) removing spontaneous quarry residents (to reduce competition with residents); (5) adding seeds of five stress-hardy exotic perennial species previously observed to thrive on quarry floors (to increase

TABLE 1. Properties of alvar species and weeds experimentally added to limestone quarry floors in southern Ontario, Canada.

Species	Family	Functional type	Alvar type?	Alvar confined? [†]	Conservation status [‡]
<i>Aquilegia canadensis</i>	Ranunculaceae	early forb	yes	...	...
<i>Asclepias syriaca</i>	Asclepiadaceae	late forb	yes	...	...
<i>Bromus kalmii</i>	Poaceae	grass	yes	50–75%	restricted
<i>Campanula rotundifolia</i>	Campanulaceae	late forb	yes	...	...
<i>Castilleja coccinea</i>	Scrophulariaceae	late forb	no	50–75%	restricted
<i>Coreopsis lanceolata</i>	Asteraceae	late forb	no	<86%	restricted
<i>Elymus trachycaulus</i>	Poaceae	grass	yes	...	...
<i>Fragaria virginiana</i>	Rosaceae	early forb	yes	...	...
<i>Geum triflorum</i>	Rosaceae	late forb	yes	<86%	restricted
<i>Hedyotis longifolia</i>	Rubiaceae	late forb	no	...	...
<i>Iris lacustris</i>	Iridaceae	early forb	no	50–75%	rare, endemic
<i>Liatris cylindracea</i>	Asteraceae	late forb	no	50–75%	restricted
<i>Penstemon hirsutus</i>	Scrophulariaceae	early forb	yes	...	...
<i>Potentilla arguta</i>	Rosaceae	late forb	yes	...	restricted
<i>Rosa blanda</i>	Rosaceae	late forb	yes	...	...
<i>Schizachyrium scoparium</i>	Poaceae	grass	no	...	...
<i>Solidago ptarmicoides</i>	Asteraceae	late forb	yes	<86%	rare
<i>Sporobolus heterolepis</i>	Poaceae	grass	yes	<86%	...
<i>Chrysanthemum leucanthemum</i>	Asteraceae	late forb	weed	...	...
<i>Daucus carota</i>	Apiaceae	late forb	weed	...	...
<i>Plantago lanceolata</i>	Platinaceae	late forb	weed	...	...
<i>Potentilla recta</i>	Rosaceae	late forb	weed	...	...
<i>Sedum acre</i>	Crassulaceae	succulent	weed	...	...

Notes: Nomenclature follows Newmaster et al. (1998). Species considered “characteristic” of alvars (natural limestone pavements) by Catling and Brownell (1995) were classified as alvar type. Confinement data and conservation status follow Catling (1995).

[†] An ellipsis indicates the species is not confined to alvars.

[‡] An ellipsis indicates the species has no recognized conservation status.

competition with weeds typical of quarries but not alvars). The sixth treatment consisted of adding just the five weed species and no alvar species; these weed species consisted of the five most abundant exotic species already present at all sites (as determined by Tomlinson et al. [2008]) and thus set a standard for species with high quarry floor colonization potential (i.e., propagule- but not microsite-limited). Fewer weed species were added than alvar species due to management concerns, but to maximize the comparability of weed and alvar assemblages, on average a greater mass of seed was added for each weed species (0.3 ± 0.2 g/species) than for each alvar species (0.19 ± 0.04 g/species), and on average commercially available weed species had greater germination rates ($53\% \pm 6\%$ of seeds/species) than alvar species ($38\% \pm 7\%$ of seeds/species) in preliminary seed viability trials (Appendix: Table A1).

Substrate amendments consisted of 2 L of sterilized silica “playground”-grade sand (Nu Gro Technologies, Brantford, Ontario, Canada) mixed with low-nutrient, pH-neutralized sphagnum peat moss (Farfard, Saint Bonaventure, Quebec, Canada), at a 7:1 ratio by fresh mass. For the fertilizer treatment, slow-release urea/formaldehyde fertilizer (38:0:0 granular “Nitroform”; Nu Gro Technologies) was intermixed with the sand/peat amendment at a rate of 16 g/plot. Substrate amendments were thoroughly mixed with the soil already present in plots, increasing plot depth by ~3–5 cm. Weed removal was accomplished by spraying plots

with 50 mL of 2% glyphosphate (RoundUp; Monsanto, St. Louis, Missouri, USA) two weeks prior to seed addition, followed by hand-weeding every three to four weeks throughout the first growing season.

Measurements and analysis

Dependent and independent variables.—Colonization success was measured at each plot for three years following seed addition by counting for each seeded species all living adult (i.e., true leaves emerged) stems or tillers present per plot and calculating the number of species (species richness, SR), number of individuals (community density, CD), and number of individuals per species (population density, PD) present per plot. Only seeded weed species were tracked in “weeds only” plots while only seeded alvar species were tracked in all other plots. The percentage of seeded species to establish (i.e., [(SR/number of species added) $\times$ 100%]; potential richness realized, PRR) was additionally calculated to enable standardized species richness comparisons among plots receiving alvar vs. weed species. Analysis of colonization success presented here is based on repeated plot sampling performed in September 2004, May 2005, August 2005, and May 2006.

Several environmental variables were measured as potential covariates of colonization success. Plots were subdivided into 18 (10 $\times$ 10 cm) subplots: at the center of each subplot soil depth was measured and the class of microhabitat present was recorded. Microhabitat classes reflect quarry surface features other than vascular plant

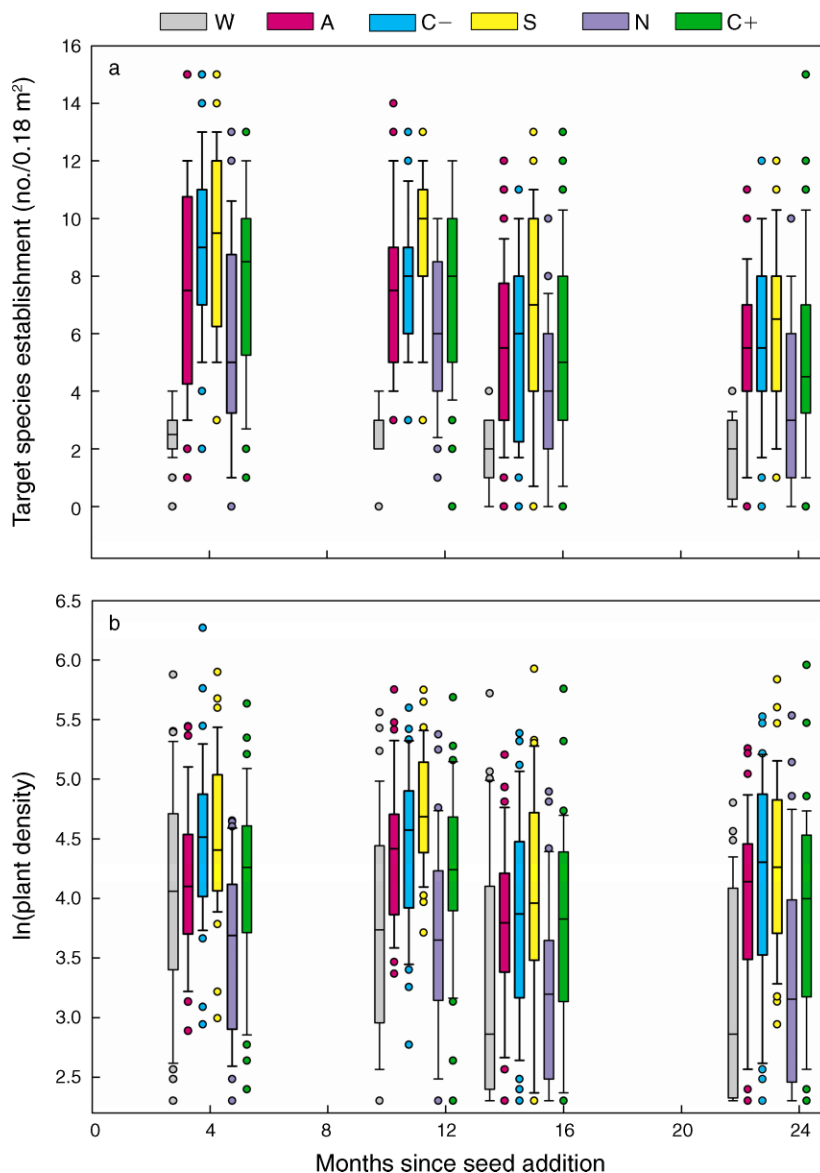


FIG. 2. Colonization success of characteristic alvar species (plants growing on natural limestone pavements) and quarry weeds over 23 months following seed addition to abandoned quarry floors. Small-plot communities ($N = 216$) established under six plot amendment treatments were monitored from June 2004 through May 2006 with respect to the number of (a) seeded species and (b) individuals ($\log_e[\text{no. plants} + 10]/0.18 \text{ m}^2$) to establish in plots. Box plots illustrate the raw sample (i.e., pre-ANCOVA) median (horizontal line), interquartile range (box), 95% confidence interval (whiskers), and outliers (circles) for each treatment at each sampling date. Letter codes correspond to treatments as follows: W, seeds of weed species added; A, seeds of alvar species added; C-, competitors removed; S, alvar seeds + silica/peat supplement added; N, alvar seeds + silica/peat + N added; C+, competitors added. In the treatment receiving alvar and weed species (C+), only responses exhibited by alvar species were measured.

cover; we classified each subplot as covered by either bare rock, till, gravel, soil, litter, woody debris, bryophytes, or lichens (definitions follow Lundholm and Larson [2003]). Data collected at the subplot scale was used to calculate at the plot scale the frequency of cover by each microhabitat, the total number of different microhabitats present (microhabitat richness, MR), mean soil depth, and the coefficient of variation for soil depth (sdCV). We expected more species might

establish in more spatially heterogeneous plots (estimated by MR and sdCV) due to reduced competitive exclusion (Lundholm and Larson 2003).

Analysis of constraints on colonization success.—Repeated-measures analysis of covariance (RM-ANCOVA) was used to separately analyze PRR and CD as functions of plot treatment, time (i.e., sampling date), the interaction between treatment and time, and environmental covariates, all blocked by site location.

TABLE 2. Repeated-measures ANCOVA results for two metrics of colonization success.

Effect	df	Potential richness realized		Total plant density	
		<i>F</i>	<i>P</i>	<i>F</i>	<i>P</i>
Treatment	5, 204	19.57	<0.0001	32.13	<0.0001
Time	3, 623	49.17	<0.0001	29.55	<0.0001
Treatment $\times$ time	15, 623	0.36	0.9880	0.81	0.6695
Site	3, 204	36.86	<0.0001	26.27	<0.0001
Moss (–)†	1, 204	5.45	<0.0001	4.71	0.0312
CV soil depth (–)†	1, 204	21.84	0.0205	0.76	0.3829

Note: Boldface indicates effects that are statistically significant ($P < 0.05$).

† Sign indicates a negative correlation.

Community density was $\log_e$ -transformed to improve normality of residuals and meet assumptions of variance analysis. To test whether alvar species had quarry colonization potential equivalent to that of typical quarry weed species, each alvar species treatment was compared to the “weeds only” treatment using Dunnett’s test for multiple comparisons to a single control. To test the influence of hypothesized microsite constraints on assembly (e.g., excessive CaCO_3 , deficient N in substrate, etc.), preplanned contrasts were performed among select pairs of treatments receiving alvar seeds. Repeated-measures analyses were conducted using Proc Mixed in SAS version 9.12 (SAS Institute, Cary, North Carolina, USA). Results reported as value ranges represent 95% confidence intervals of least-square means (back-transformed in the case of CD).

Population-level colonization success was evaluated by calculating for each seeded species both the proportion of seeded plots supporting more than one individual per plot per year (frequency of establishment, FOE), and average PD ($\log_e$ -transformed) in the subset of plots where establishment had occurred (density where established, DWE). Each metric was analyzed separately using ANOVA, and Tukey-Kramer-adjusted multiple comparisons were used to contrast species performances. Frequency of establishment was replicated at the site scale and modeled as a function of species identity alone, while DWE was replicated at the plot scale and modeled as a function of species identity blocked by site. Given the unequal proportion of each species in the seed mix, species comparisons were primarily performed to evaluate different contributions to community patterns rather than to determine which species performed “best.”

Comparison to alvar reference communities.—We investigated the influence of alvar species addition on quarry floor communities by surveying both experimentally introduced and spontaneously present species in a subset of alvar-seeded plots ($N = 38$) and in a set of unseeded control plots ($N = 22$) located at one site and sampled in June 2006. Seeded plots consisted of all plots on-site that had received alvar seeds and supported >10 individuals as of May 2006. Control plots were of equal dimensions to seeded plots and placed at random locations that met criteria of $<50\%$ bare rock cover

and $>30\%$ vegetation cover. Both seeded and unseeded quarry plots were compared to untouched reference plots located on a pristine Bruce Peninsula alvar of high conservation priority (Pendall Point; $45^\circ 11' 08.3''$ N, $81^\circ 35' 58.0''$ W), using previously collected alvar data (see Lundholm and Larson 2003). Of the 144 randomly located, 0.3×0.6 m alvar plots for which data was available, we included the 38 most species-rich plots (with respect to vascular plants) in comparative analysis to represent conservative “best case” endpoints for quarry-to-alvar rehabilitation. At all plots the total number of vascular plant species and the frequency of vegetation cover (based on presence/absence of living plant biomass in 18 subplots per plot) were measured, and these properties were compared among plot types (i.e., seeded quarry, unseeded quarry, or alvar) using ANOVA. Regression analysis was performed using data from seeded quarry plots to determine whether colonization by alvar species was related to quarry resident species richness or vegetation cover.

RESULTS

Propagule limitation

While the success of quarry floor colonization by seeded assemblages varied greatly among plots, 214 of 216 seeded plots supported establishment of at least one species by the end of the first growing season, and 197 plots still supported introduced populations by the third growing season. At each plot 60 ± 60 individuals representing $40\% \pm 20\%$ of introduced species established (raw sample means $\pm$ SD), each species established in at least two plots, and a maximum of 15 alvar species or four weed species co-occurred in a single plot. Inspection of boxplots summarizing raw establishment data in each treatment and at each sampling date revealed that colonization success was highly variable over time, within treatments, and among treatments, but differences among treatments were consistent over time, and in general more species and more individuals established in plots seeded with alvar species compared to those seeded with weed species (Fig. 2).

Variance in PRR was significantly explained by effects of treatment, time, site, and two environmental variables, but not by interaction between treatment and time (Table 2). On average 45% (42–47%) of seeded weed

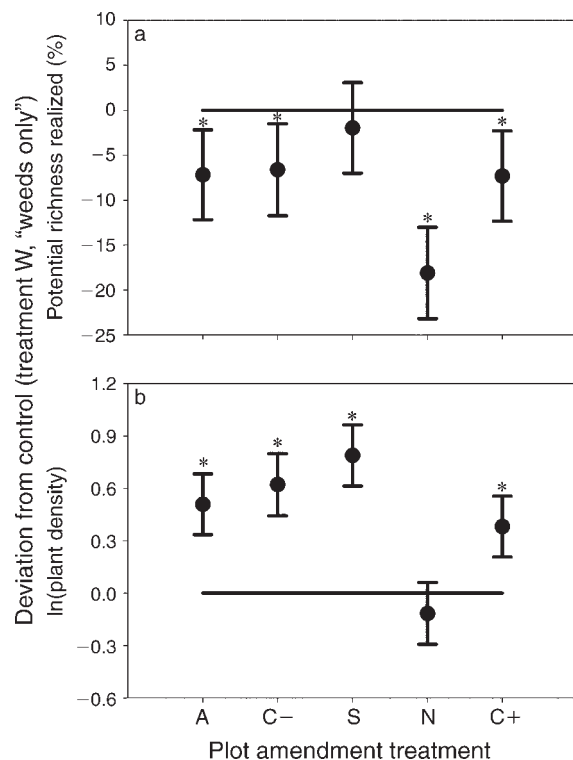


FIG. 3. Deviation in colonization success exhibited by alvar assemblages under different treatment regimes compared to the assemblage of common quarry weeds. Symbols represent least-square means and 95% confidence intervals for differences between each alvar-seeded treatment and the weed-seeded control treatment; significant deviation (determined using Dunnett-Hsu adjusted multiple-comparison tests $P < 0.05$) is indicated by an asterisk. The horizontal bar is the zero line. Estimated deviation accounts for mean differences across all sampling dates, given the absence of a significant treatment $\times$ time interaction in repeated-measures ANCOVA. Colonization success was estimated by (a) the percentage of seeded species to become established per plot and (b) the total number of seeded individuals to establish per plot ($\log_e[\text{no. plants} + 10]/0.18 \text{ m}^2$). Significant differences in the degree of deviation exhibited by different alvar-seeded treatments are shown in Table 3. In the treatment receiving alvar and weed species (C+), only responses exhibited by alvar species were measured. See Fig. 2 for an explanation of treatment abbreviations.

species established per plot in the “weeds only” control treatment. The percentage of seeded alvar species to establish in each of the other treatments was lower than this value, but in most cases the magnitude of deviation was small and only one treatment supported establishment of $<37\%$ of seeded species per plot (Fig. 3a). The difference between PRR in weed-seeded vs. alvar-seeded plots was not significant in plots receiving the low-nutrient substrate amendment, but was significant for each of the other alvar species treatments (Table 3). Variance in CD was similarly explained by effects of treatment, time, site, and an environmental variable (Table 2), but in contrast to PRR, four of the five alvar species treatments exhibited significantly greater CD than did the “weeds only” treatment (e.g., 22–31 plants/

plot in which only weeds were added vs. 43–58 plants/plot in which only alvar species were added; Fig. 3b, Table 2). Taken together these patterns indicate strong propagule limitation of quarry floor colonization by both hardy weeds of known high-colonization potential and native rock outcrop specialists characteristic of alvars.

Microsite constraints

Potential microsite limitations on quarry colonization by alvar species were investigated by comparing plots seeded with alvar species that received or did not receive various microsite alterations and by identifying environmental features that explained variance in colonization success. Plots receiving alvar species exhibited greater PRR and CD when seed addition was accompanied by the low-nutrient substrate amendment (40–45% PRR, 60–80 plants/plot CD) than when alvar seeds alone were added (35–40% PRR, 43–58 plants/plot CD), but PRR and CD dropped sharply where the high-nutrient substrate amendment was applied (24–30% PRR, 18–26 plants/plot CD; Table 3). Quarry colonization by alvar species was thus not limited by substrate N deficiency but concentrated limestone tailings in quarry substrate moderately constrained community assembly. In contrast, neither PRR nor CD differed significantly between plots receiving alvar species only vs. plots receiving seeds of both alvar and weed species or between plots receiving alvar species only vs. plots receiving seeds of alvar species and removal of all vegetation other than seeded alvar species. However, CD in plots in which potential competitors were removed (49–66 plants/plot) was significantly greater than in plots to which they were added (36–49 plants/plot; Table 3). Competition with spontaneously occurring quarry resident species thus did not limit colonization by alvar species except under maximal abundance of weeds relative to alvar species.

Of 12 environmental features analyzed, only two factors explained significant variance in PRR or CD, and this variance was small ($<2\%$ of total variance) relative to that explained by other factors ($\sim 30\%$ of

TABLE 3. Microsite composition and soil depth in plots.

Environmental variable	Mean	SD
Mean depth (cm)	5	5
CV depth	46	20
Microhabitat richness	4	1
Soil (%)	31	25
Litter (%)	30	22
Moss (%)	18	26
Gravel (%)	10	13
Till (%)	6	9
Lichen (%)	3	9
Woody debris (%)	1	4
Rock (%)	1	3

Note: Values represent sample means and standard deviations calculated for plots ($N = 216$), based on data collected within subplots ($N = 18$ per plot).

TABLE 4. Contrasts used to test colonization constraint hypotheses.

Contrasts performed	Richness realized (%)			Community density		
	<i>t</i>	<i>F</i>	<i>P</i>	<i>t</i>	<i>F</i>	<i>P</i>
Dispersal distance limitations						
W vs. A	-3.64		0.0016	5.78		<0.0001
W vs. S	-0.98		0.786	8.87		<0.0001
W vs. N	-9.01		>0.0001	-1.29		0.5701
W vs. C-	-3.28		0.0056	6.90		<0.0001
W vs. C+	-3.69		0.0014	4.31		0.0001
Soil quality						
A vs. S		6.68	0.0104		9.60	0.0022
S vs. N		66.47	<0.0001		104.72	<0.0001
Competition with quarry residents						
A vs. C-		0.08	0.7834		1.55	0.215
A vs. C+		0.00	0.9483		2.07	0.1521
C- vs C+		0.12	0.7326		7.16	0.0081

Notes: Letter codes correspond to treatments as follows: W, seeds of weed species added; A, seeds of alvar species added; S, alvar seeds + silica/peat supplement added; N, alvar seeds + silica/peat + N added; C-, competitors removed; C+, competitors added. Boldface indicates statistically significant contrasts ($P < 0.05$). Multiple comparisons with control treatment (W) were performed using the Dunnett-Hsu adjusted *t* test ($df = 204$ for all), while preplanned contrasts among specific alvar-seeded treatments were performed using *F* tests ($df = 1, 204$ for all).

total variance was explained by all factors combined). Potential richness realized increased slightly with soil depth homogeneity, and both PRR and CD increased slightly with decreasing bryophyte cover. For each environmental variable plot-scale means were associated with large standard deviations, indicating high spatial environmental heterogeneity on quarry floors (Table 4).

Influence of alvar species addition on quarry communities

Unseeded quarry floor plots supported communities of 9–12 (maximum = 17) vascular species per 0.18 m² plot while plots seeded with alvar species supported 17–20 (maximum = 28) vascular species, about half of which were introduced alvar natives (Fig. 4a). In contrast, maximum-diversity alvar reference plots supported communities of 20–23 (maximum = 30) vascular species per 0.18 m². Plot type significantly explained 49% of the variance in species richness ($F_{2,95} = 46.14$, $P < 0.0001$). Differences in mean species richness among plot types were statistically significant ($P < 0.0001$ for seeded vs. unseeded quarry plots; $P = 0.02$ for seeded quarry vs. unseeded alvar plots). In contrast, mean vegetation cover was variable but generally high within each type of plot and statistically indistinguishable among plot types ($P > 0.17$; Fig. 3b). The success of alvar assemblages seeded to quarry plots was unrelated to the diversity or abundance of quarry residents already present in plots ($P > 0.15$).

Individual species responses

Species identity had a strong influence on both the frequency and density with which individual populations established, but there was no obvious pattern of separation among the seeded weed and alvar species (Appendix: Table A2). Top-performing species included the weedy forb *Daucus carota* and the restricted, alvar-

confined grass *Bromus kalmii*, each establishing in >90% of the plots to which they were seeded with densities of 11–19 individuals/plot. The rare and highly confined alvar forb *Solidago ptarmicoides* established frequently, as did typical alvar species *Asclepias syriaca* and *Schizachyrium scoparium* and the hardy weed *Plantago lanceolata*, each colonizing >60% of seeded plots and achieving average densities of >4 individuals/plot. The rare *Iris lacustris*, for which only five seeds per plot were available for planting, established uncommonly but distinctly, with 13 individuals established among 10 plots and three sites by the third growth season.

DISCUSSION

Colonization of long-abandoned limestone quarry floors by species assemblages typical of naturally occurring limestone pavements (alvars) was strongly propagule-limited and weakly microsite-limited. Communities that had spontaneously established on patches of quarry floor expressed dramatically increased species richness following addition of alvar species, resembling natural alvar communities of high conservation value. Given that quarries represent a wide class of degraded ecosystems (i.e., those that are highly stressed and resistant to conventional restoration methods), while alvars represent natural high-stress ecosystems closely analogous to such sites, strong dispersal but not microsite limitations on quarry colonization by alvar assemblages indicates targeting DSA ecosystems as reference points in rehabilitation may provide a novel, general approach to efficiently meeting both conservation and reconstruction goals. Conceptually, niche differentiation resulting from environmental selection over both ecological and evolutionary timescales explains the success of the DSA reconstruction model and

predicts high potential for its utility in a variety of ecosystems.

Constraints on the colonization of quarry floors by alvar species

The results of many previous seed addition experiments suggest that colonization of landscapes by biotic communities is generally controlled by varying combinations of propagule/dispersal limitations, microsite limitations, and species interactions, with the precise balance of factors often dependent upon environmental context (e.g., Tilman 1997, Zobel 1997, Turnbull et al. 2000, Foster 2001). In the context of rehabilitating severely degraded ecosystems, the capacity for species to tolerate stressful microsites may be more important than propagule limitation or competition (Ash et al. 1994, Bakker and Berendse 1999, Khater et al. 2003, Zelnik et al. 2008). Niche theory posits that over ecological time harsh abiotic conditions select for establishment of species with functional traits conferring stress tolerance, while over evolutionary time these same stressors select for morphological and behavioral adaptations that ultimately improve the “fit” of organisms to their environment (Ackerly 2003, Agosta and Klemens 2008). However, as managers of damaged ecosystems are generally unwilling to wait for evolution before obtaining appropriate species to use in reconstruction, we hypothesized that species that have evolved under ecological conditions analogous to the DS may already be adapted to microsite constraints limiting ecosystem recovery and predicted such species would be primarily propagule-limited under DS microsite conditions that prevent establishment of many alternative assemblages. This idea is similar to but distinct from “preadaptations” explaining the success of some invasive species, in which ancestral traits become co-opted to perform new, fitness-enhancing functions in novel environments (exaptation; Ackerly 2003, Mack 2003). We argue that from the point of view of DSA species, the DS environment is not novel at all, but ecologically analogous to the ancestral environment such that traits conferring fitness in the DSA similarly confer fitness in the DS.

Abandoned limestone quarry floors seeded with alvar species demonstrated the pattern predicted by the DSA hypothesis exactly. Sites that had remained mostly barren despite 30–70 years of abundant seed rain from surrounding ecosystems were successfully colonized by species from biophysically equivalent alvar ecosystems following seed addition alone. Though not every seeded species emerged in every plot (or indeed, in any single plot), each species established in some plots, >99% of plots supported some species, and the introduced alvar assemblage performed approximately as well as the assemblage of species with known high quarry colonization potential; taken together these results indicate that spatially variable microsite constraints exist, but quarry floors are more generally seed-limited with

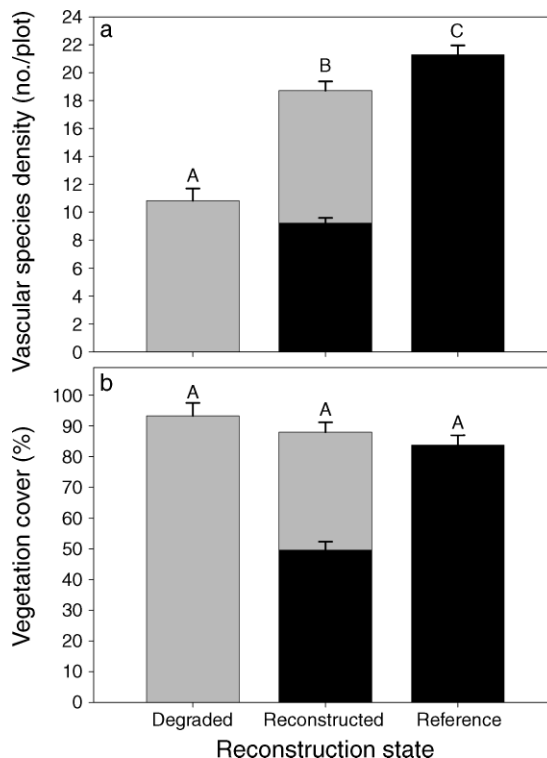


FIG. 4. Community properties in degraded, reconstructed, and reference limestone pavements. Unseeded quarry plots (“Degraded”), quarry plots seeded with 18 alvar species (“Reconstructed”), and plots located on a natural alvar pavement (“Reference”) were compared with respect to (a) the total number of vascular plant species present (including nontarget species on quarry floors) and (b) the percentage of plot surface area covered with living vegetation. Black portions of bars represent alvar species while gray portions represent species that have spontaneously colonized quarry floor plots. Least-square means and standard errors are shown for each restoration state; bars with shared uppercase letters are not significantly different from one another according to Tukey-Kramer adjusted multiple comparison tests ($P < 0.001$ for difference in richness between seeded and unaltered quarry plots; $P = 0.02$ for difference in richness comparison between seeded quarry and reference alvar plots; $P > 0.17$ for all contrasts of vegetation cover values).

respect to alvar species. Likewise, while increased species richness and community density resulting from the low-nutrient substrate amendment indicated properties of the quarry substrate somewhat limited the assembly of alvar communities, the low magnitude of these increases (e.g., 13% more species and 40% more individuals established on average where substrate was amended) confirms that microsite constraints on alvar species are weak relative to seed limitation. This is consistent with alvar species being physiologically suited to tolerate harsh quarry conditions, but constrained from colonizing open sites spontaneously due to dispersal barriers such as great distances between relatively small quarry and alvar “islands.” The small benefit obtained from substrate amendment likely accrued due to dilution of

limestone extraction tailings comprising much of the quarry substrate, as peat moss was included primarily to counter effects of sand addition on substrate bulk density and organic matter and increased substrate depth was generally unrelated to colonization success. Excessive CaCO_3 may have negative effects on plant growth by reducing the bioavailability of phosphorus, iron, and other nutrients (Brady 1974) or by coating plant dermal tissue during flood events and damaging cuticle integrity upon drying, thereby increasing sensitivity to drought (Dixon and Hamblen 1988). In contrast, negative effects of nitrogen addition are inconsistent with nitrogen-deficient quarry substrate limiting colonization success.

While competitive species interactions have been a major focus of research on community assembly, we found no evidence that competition between establishing alvar species and quarry resident species at typical densities limited the success of alvar species. The practical implication of these results for quarry managers is the recommendation to focus rehabilitation strategies on introducing suitable alvar species while making minimal substrate amendments, rather than diverting resources toward weed removal, extensive soil addition, or other site alterations. The theoretical implication is that because a typical severely damaged environment was primarily propagule-limited with respect to a typical DSA assemblage, targeting DSA ecosystems as reconstruction reference systems may provide effective tools for generally improving reconstruction success in challenging environments.

Biodiversity and the conservation significance of DS-DSA ecosystems

Quarry floors colonized by introduced alvar plant species exhibited high biodiversity at fine spatial scales, with species richness approximating that observed in maximum-diversity reference plots located on a natural alvar. While this reference site is federally protected due to extreme biodiversity and refuge provided to rare and threatened species, seeded quarry plots also provided refuge to species of conservation concern, including *Iris lacustris*, *Solidago ptarmicoides*, *Bromus kalmii*, and *Liatriis cylindracea* (see Table 1). Reasons for preserving biodiversity have been argued extensively, drawing upon biodiversity's potential roles in determining the magnitude and variability of ecosystem process rates, providing ingenuity, genetic novelty, and disease resistance in biological systems used by humans and fulfilling human psychological needs. Less clear is why some locations are extraordinarily biologically diverse compared to others ("biodiversity hot spots") or how high diversity can be built into human-created ecosystems. Though unproductive, natural high-stress ecosystems such as serpentine, barrens, and rock outcrops are frequently biodiversity hot spots, and our results suggest that heavily damaged anthropogenic wastelands have similar potential.

Several potential explanations for high-diversity, high-stress ecosystems in nature exist: similarity among species may not be limiting when populations are constrained more by within-species stress responses than among-species competition (Leibold and McPeck 2006); high stress may favor positive species interactions that expand the realized niches of stress-intolerant species into harsh environments via facilitation by stress-tolerant species (Michalet et al. 2006); harsh environments may be associated with particularly high spatial and temporal heterogeneity that prevents competitive exclusion (Lundholm and Larson 2003, Freestone and Inouye 2006); abiotic extremes may select for evolutionary adaptations conferring stress tolerance, a niche-differentiation process possibly accelerated by the "island"-like properties of many high-stress ecosystems that are discontinuous with the surrounding landscape (Ackerly 2003). Understanding which mechanisms are responsible for coexistence in high-stress ecosystems may inform management strategies for "packing" diverse assemblages into anthropogenic ecosystems; however, the DSA reconstruction approach is pragmatic even in the absence of such detailed understanding, as it utilizes species and communities that have already interacted with and adapted to DS-like conditions, by whatever mechanism. Furthermore, high-stress ecosystems not associated with high local biodiversity are often inhabited by regionally uncommon or endemic specialist species, and thus replicating such ecosystems in DS sites has potential to greatly increase regional as well as local biodiversity (Freestone and Inouye 2006, Esfeld et al. 2008).

Applying the DSA model to reconstruction and conservation management

Our results suggest that two main benefits emerge from targeting DSA ecosystems while managing severely damaged landscapes. First, sites that are extremely difficult to restore because of stringent abiotic filters may be successfully and efficiently colonized by species that naturally fit the "mesh" of such filters (due to their evolutionary life history under analogous conditions), negating the need to mechanically alter filter structure via expensive site engineering. Secondly, DSA ecosystems targeted in such efforts can themselves be valued conservation targets due to their rarity, threatened status, and function as refugia for high biodiversity; extension of DSA assemblages into analogous degraded environments may thus provide critical ex situ conservation to species at risk of extinction due to destruction of their uncommon natural habitats. The degree of similarity between DS and DSA ecosystems may be pivotal to the success of the DSA model; where the DS is strongly analogous to a natural ecosystem, only dispersal limitations will need to be overcome; however, where anthropogenic ecosystems exhibit properties intermediary between multiple possible DSA ecosystems, many different strategies and outcomes may be

possible. Incorporation of the DSA approach into land management demands careful consideration of the value of prospective vs. retrospective restoration goals (Rosenzweig 2003, Choi 2007, Seastedt et al. 2008), while optimizing the DSA model to maximally increase the functional and conservation importance of anthropogenic wastelands will require a finer understanding of mechanisms underlying species niche relations and coexistence in harsh environments, and metacommunity dynamics among DS and DSA sites. If successful, however, such conceptual advances could dramatically increase the capacity for humans to reconcile necessary natural resource exploitation with the conservation and reconstruction of diverse, functional ecosystems.

ACKNOWLEDGMENTS

This work was funded by the National Science and Research Council (NSERC) and the Ontario Aggregate Resource Corporation. Thanks to L. Klatt, M. Bell, T. Wilson, K. Kuntz, M. Gartshore, K. Pidgin, D. Sterrett, S. Tomlinson, U. Matthes, and all site owners for assistance in completing this research; thanks to J. Klironomos, A. MacDougall, K. McCann, U. Matthes, and two anonymous reviewers for helpful comments on this manuscript.

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APPENDIX

Properties of seeds added to quarry floors and multiple comparisons among seeded alvar species with respect to frequency of establishment and density where established (*Ecological Archives* A020-022-A1).